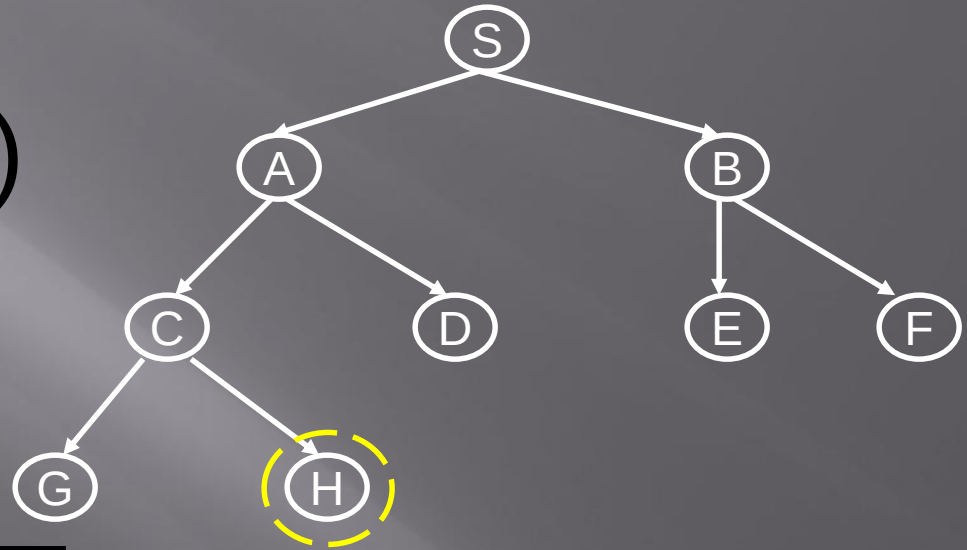


Uninformed Searches

Breadth First Searching (BFS) in AI

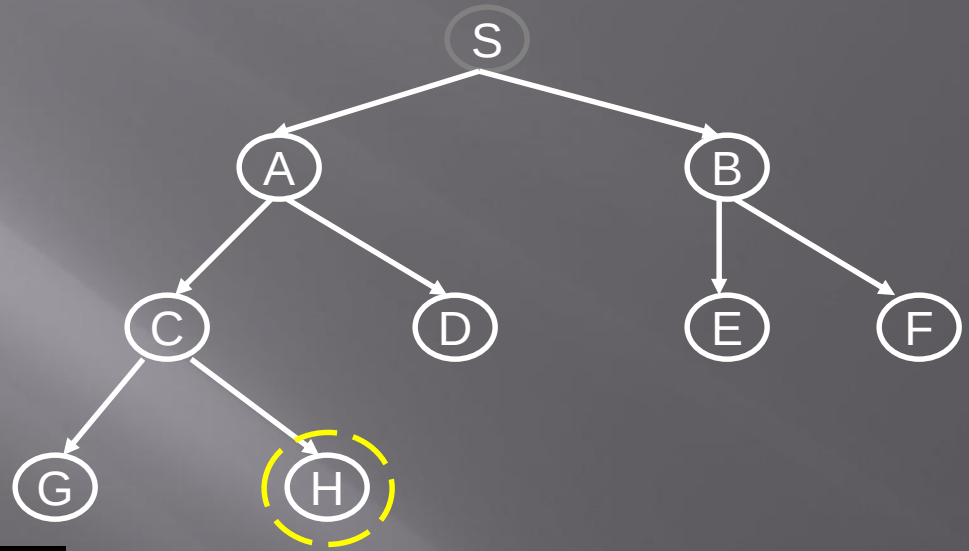
BFS: Example

$P(n) = \text{height}(n)$



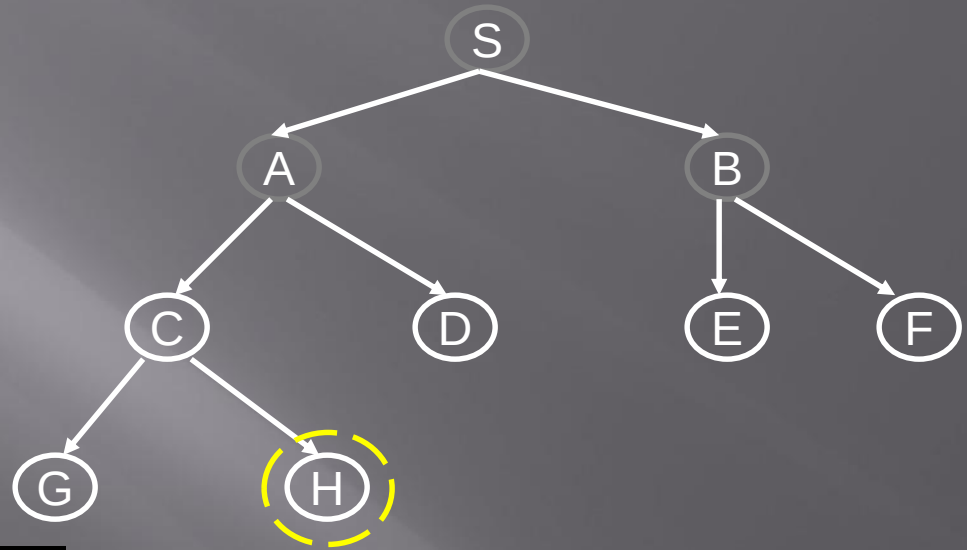
	Q	Visited
1		
2		
3		
4		
5		

BFS: Example



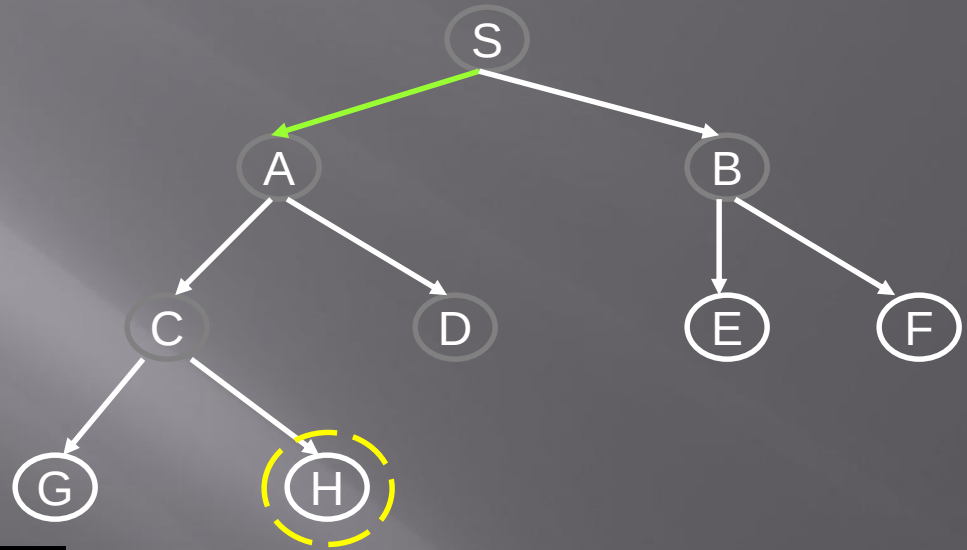
	Q	Visited
1	S	S
2		
3		
4		
5		

BFS: Example



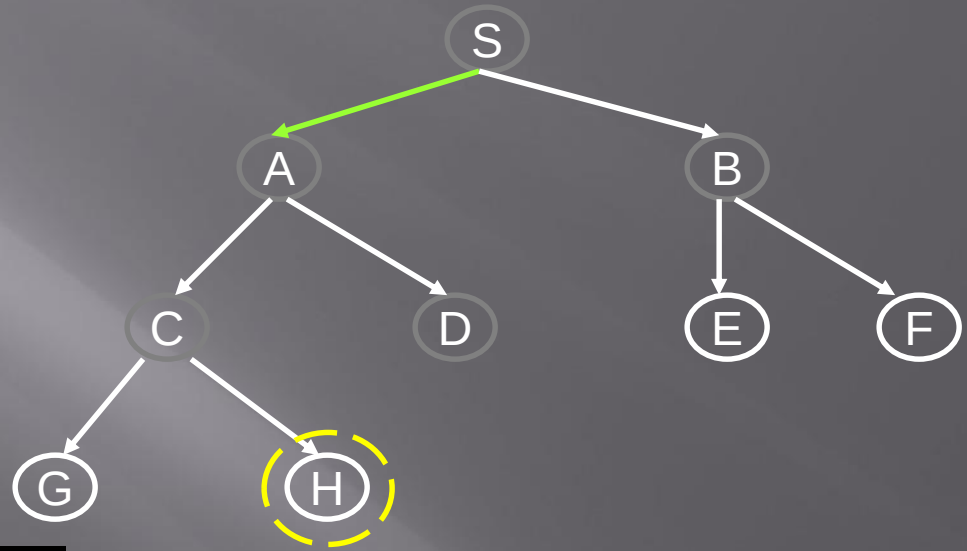
	Q	Visited
1	S	S
2	A,B	S,A,B
3		
4		
5		

BFS: Example



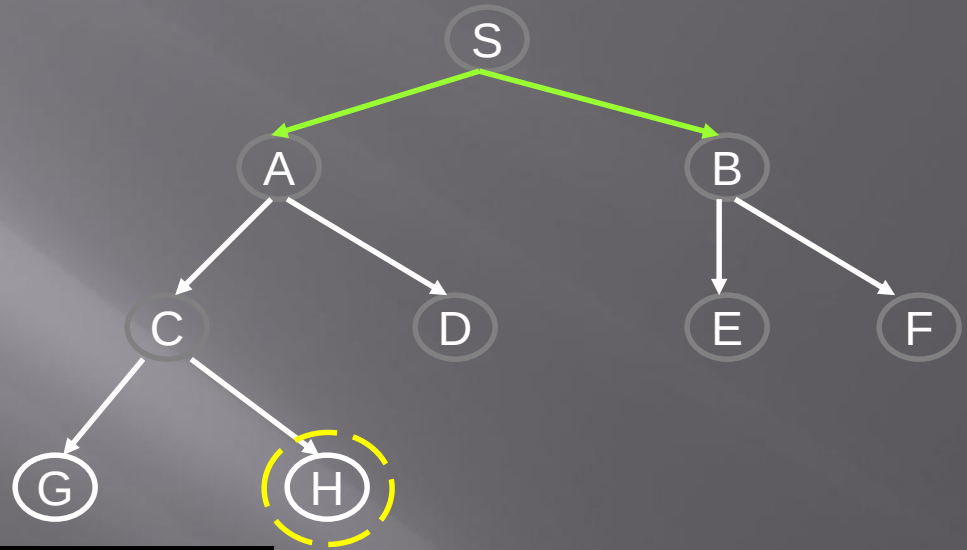
	Q	Visited
1	S	S
2	A,B	S,A,B
3	B,C,D	S,A,B,C,D
4		
5		

BFS: Example



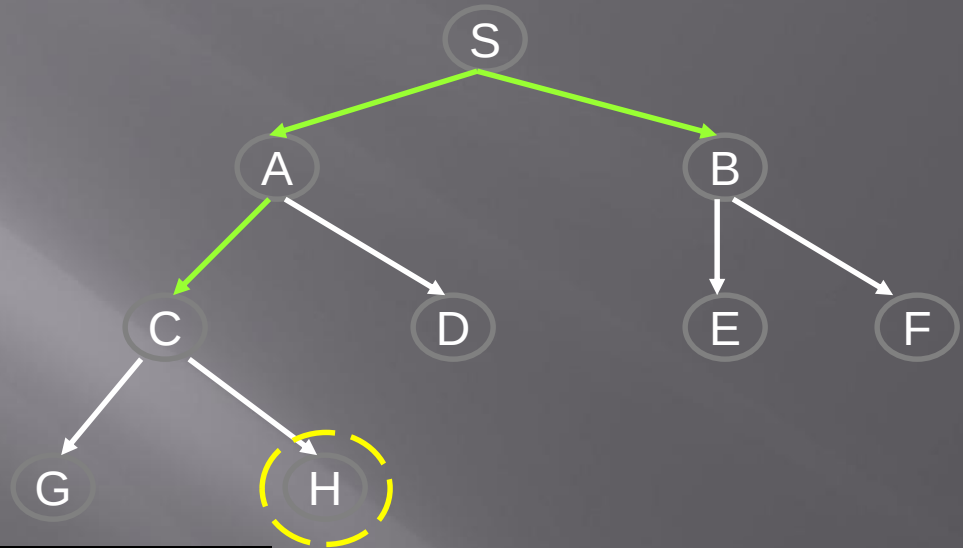
	Q	Visited
1	S	S
2	A,B	S,A,B
3	B,C,D	S,A,B,C,D
4		
5		

BFS: Example



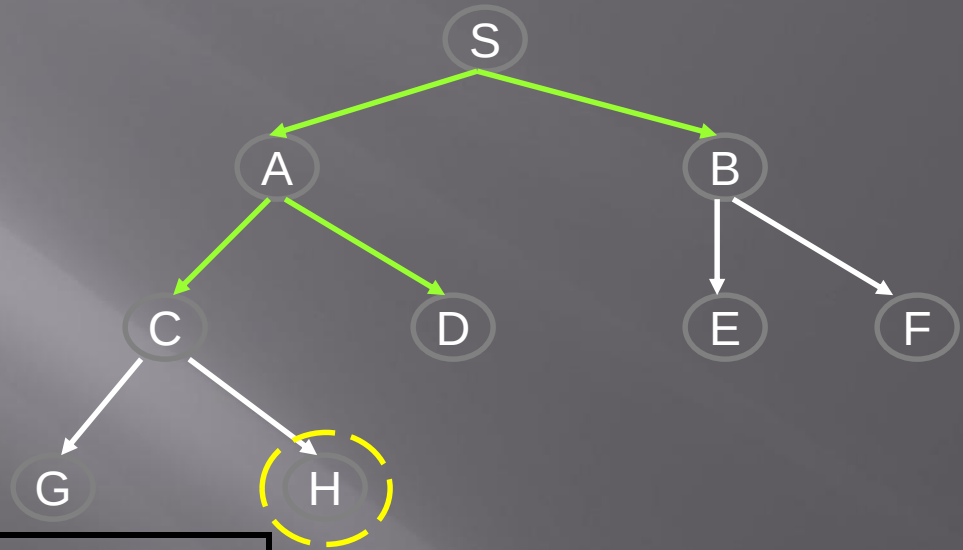
	Q	Visited
1	S	S
2	A,B	S,A,B
3	B,C,D	S,A,B,C,D
4	C,D,E,F	S,A,B,C,D,E,F
5		

BFS: Example



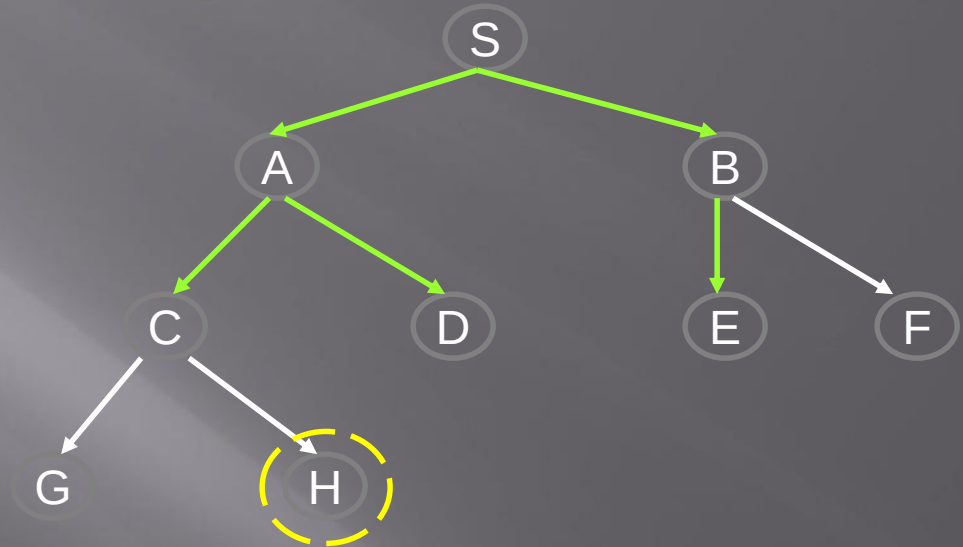
	Q	Visited
2	A,B	S,A,B
3	B,C,D	S,A,B,C,D
4	C,D,E,F	S,A,B,C,D,E,F
5	D,E,F,G,H	S,A,B,C,D,E,F,G,H
6		

BFS: Example



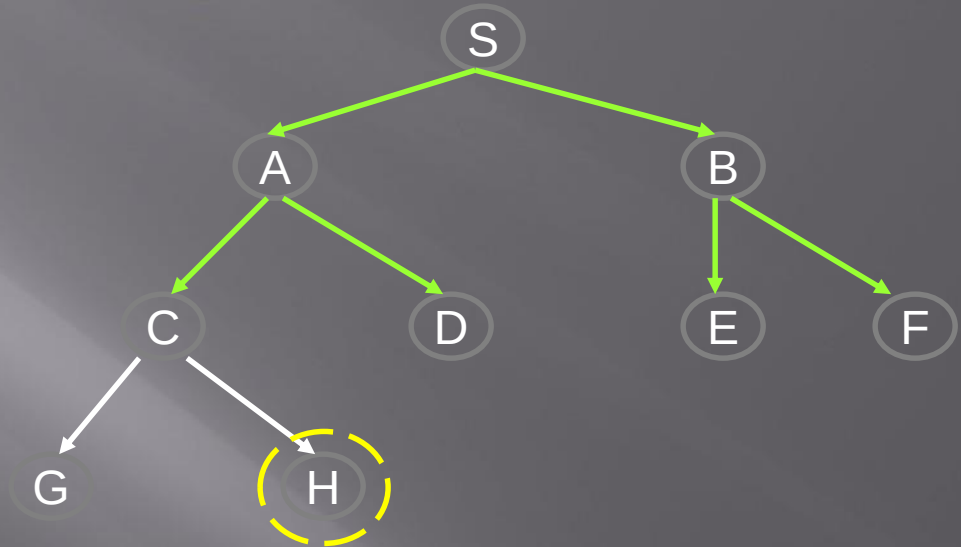
	Q	Visited
3	B,C,D	S,A,B,C,D
4	C,D,E,F	S,A,B,C,D,E,F
5	D,E,F,G,H	S,A,B,C,D,E,F,G,H
6	E,F,G,H	S,A,B,C,D,E,F,G,H
7		

BFS: Example



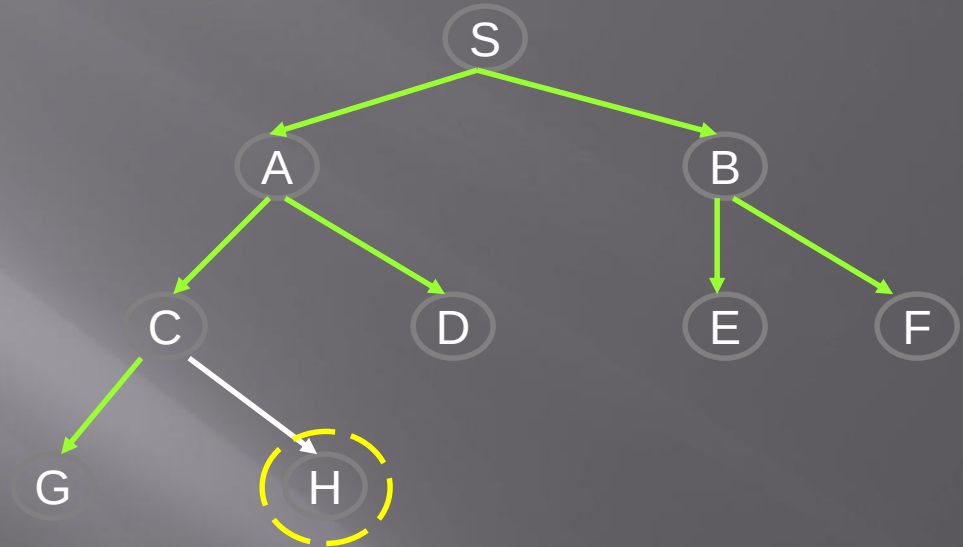
	Q	Visited
4	C,D,E,F	S,A,B,C,D,E,F
5	D,E,F,G,H	S,A,B,C,D,E,F,G,H
6	E,F,G,H	S,A,B,C,D,E,F,G,H
7	F,G,H	S,A,B,C,D,E,F,G,H
8		

BFS: Example



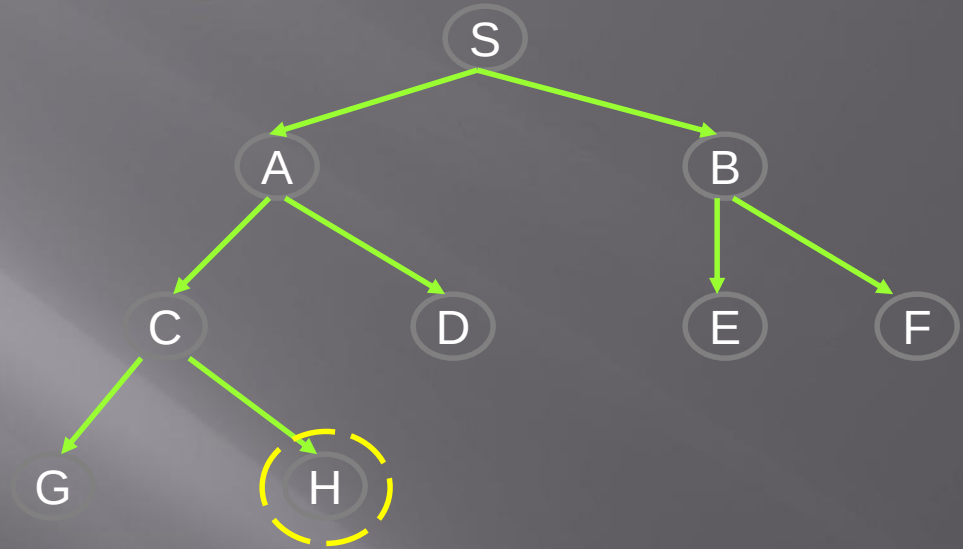
	Q	Visited
5	D,E,F,G,H	S,A,B,C,D,E,F,G,H
6	E,F,G,H	S,A,B,C,D,E,F,G,H
7	F,G,H	S,A,B,C,D,E,F,G,H
8	G,H	S,A,B,C,D,E,F,G,H
9		

BFS: Example



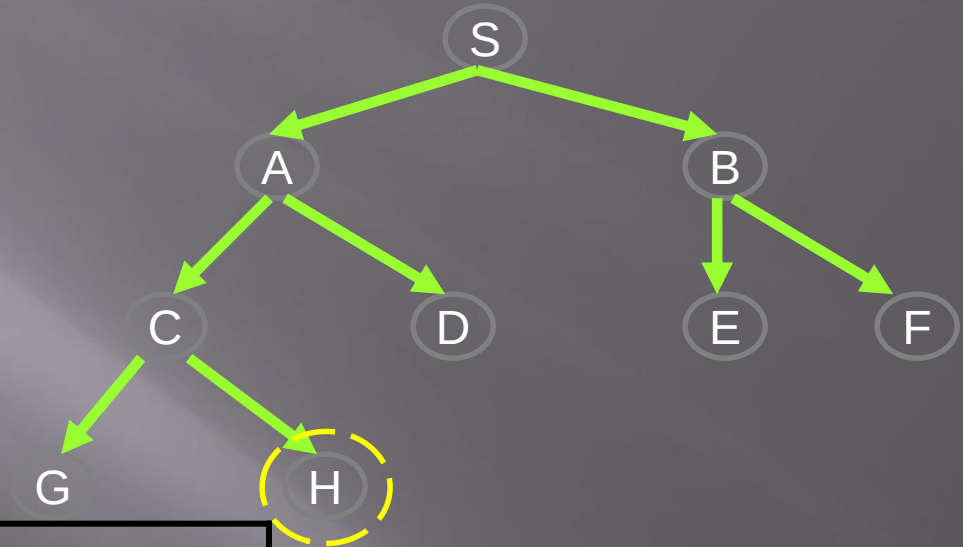
	Q	Visited
6	E,F,G,H	S,A,B,C,D,E,F,G,H
7	F,G,H	S,A,B,C,D,E,F,G,H
8	G,H	S,A,B,C,D,E,F,G,H
9	H	S,A,B,C,D,E,F,G,H
10		

BFS: Example



	Q	Visited
6	E,F,G,H	S,A,B,C,D,E,F,G,H
7	F,G,H	S,A,B,C,D,E,F,G,H
8	G,H	S,A,B,C,D,E,F,G,H
9		S,A,B,C,D,E,F,G,H
10		S,A,B,C,D,E,F,G,H

BFS: Example



	Q	Visited
1	S	S
2	A,B	S,A,B
3	B,C,D	S,A,B,C,D
4	C,D,E,F	S,A,B,C,D,E,F
5	D,E,F,G,H	S,A,B,C,D,E,F,G,H
6	E,F,G,H	S,A,B,C,D,E,F,G,H
7	F,G,H	S,A,B,C,D,E,F,G,H
8	G,H	S,A,B,C,D,E,F,G,H
9	H	S,A,B,C,D,E,F,G,H
10		S,A,B,C,D,E,F,G,H

Problem with BFS

- Imagine searching a tree with branching factor 8 and depth 10. Assume a node requires just 8 bytes of storage. The breadth first search might require up to:

$$= (8)^{10} \text{ nodes}$$

$$= (2^3)^{10} \times 2^3 = 2^{33} \text{ bytes}$$

$$= 8,000 \text{ Mbytes}$$

$$= 8 \text{ Gbytes}$$

Problems with DFS and BFS

DFS has small space requirements (linear in depth) but has major problems:

- DFS can run forever in search spaces with infinite length paths
- DFS does not guarantee finding the shallowest goal

BFS guarantees finding the lowest path even in presence of infinite paths, but it has one great problem

- BFS requires a great deal of space (exponential in depth)